

The Privacy of Synthetic Data

Eleven papers that define what a differentially private data release should promise — and where each one stops short.

SynthProof – synthetic data that ships with its proof

11

PAPERS SURVEYED

2022–2026

USENIX · VLDB · CCS · PoPETs · ICLR · CSCW

6

RESEARCH ANGLES

1

GAP WE FILL

● Attacks ● Mechanism audits ● Preprocessing audits ● Generative mechanisms
● Real deployment ● Standards & human factors

The eleven papers at a glance

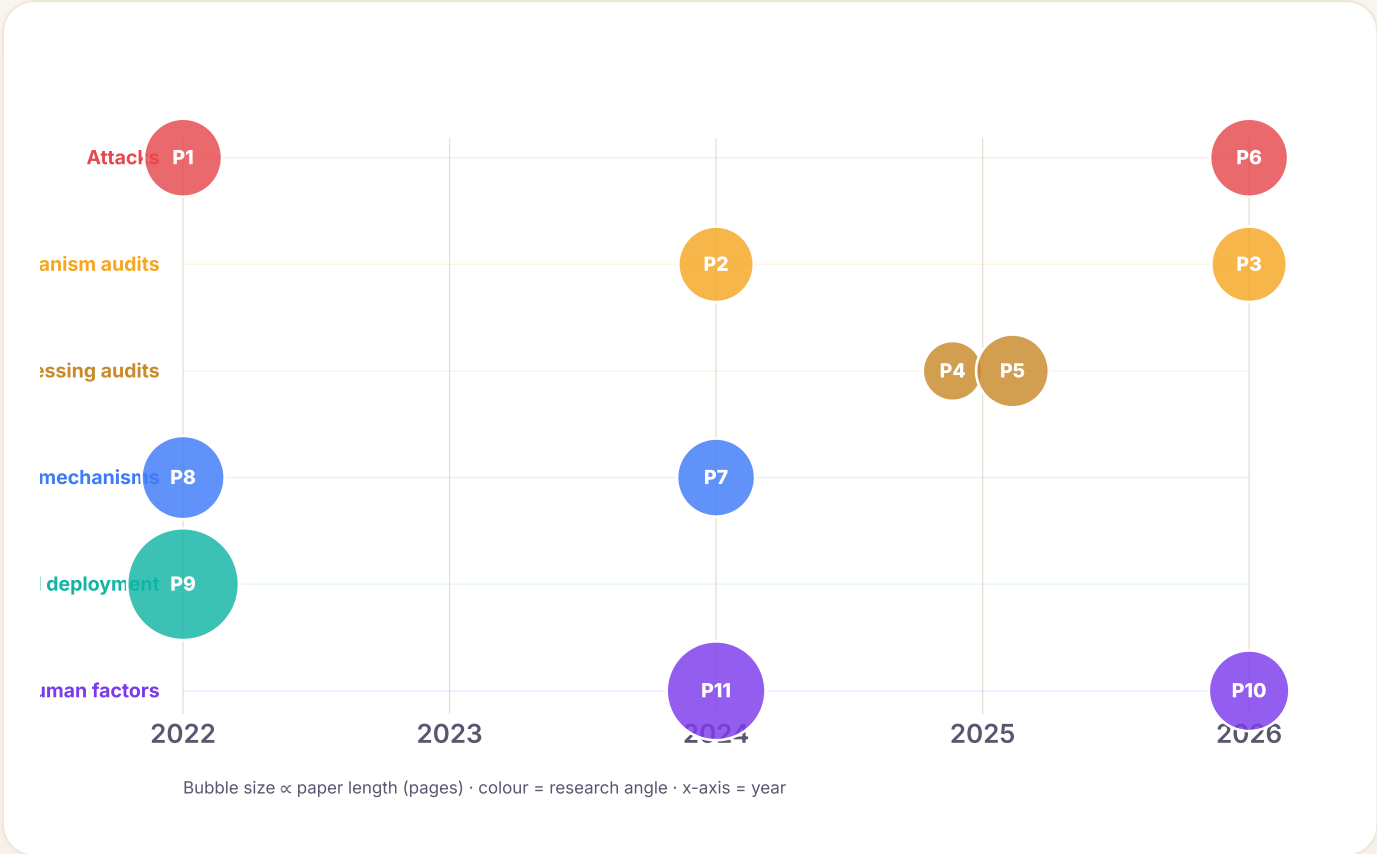
Every paper read in full and verified against its PDF — author, title, year, venue, research angle, and the datasets it used.

#	TITLE & AUTHORS	YEAR	VENUE	ANGLE	DATASETS
P1	Synthetic Data — Anonymisation Groundhog Day Stadler, Oprisanu & Troncoso	2022	USENIX Security	Attacks	Adult, Texas
P2	“What do you want from theory alone?” Tight Auditing of DP-SDG Annamalai, Ganev & De Cristofaro	2024	USENIX Security	Mechanism audits	Adult, SF Fire
P3	Privacy in Theory, Bugs in Practice: Grey-Box Auditing of DP Libraries Cebere, Erb, Desfontaines, Bellet & Fitzsimons	2026	PoPETs	Mechanism audits	—
P4	Understanding the Impact of Data Domain Extraction on Synthetic-Data Privacy Ganev, Annamalai, Mahiou & De Cristofaro	2025	ICLR SynthData	Preprocessing audits	Wine
P5	The Importance of Being Discrete: Discretization in End-to-End DP Ganev, Annamalai, Mahiou & De Cristofaro	2025	ACM CCS	Preprocessing audits	Adult, Gas, Wine
P6	SMOTE and Mirrors: Privacy Leakage from Synthetic Minority Oversampling Ganev, Nazari, Davison ... De Cristofaro	2026	ICLR	Attacks	8 imbalanced, cardio, creditcard, higgs
P7	Differentially Private Data Generation with Missing Data Mohapatra, Zong, Kerschbaum & He	2024	VLDB	Generative mechanisms	Adult, Bank, BR2000, National
P8	AIM: An Adaptive and Iterative Mechanism for DP Synthetic Data McKenna, Mullins, Sheldon & Miklau	2022	VLDB	Generative mechanisms	Adult, salary, msnbc, fire, nltcs, titanic
P9	The 2020 Census Disclosure Avoidance System TopDown Algorithm Abowd et al. (U.S. Census Bureau)	2022	Harvard Data Sci. Review	Real deployment	Census (restricted)
P10	“We Need a Standard”: Toward an Expert-Informed Privacy Label for DP Dibia, Lu, Bhattacharjee, Near & Feng	2026	PoPETs	Standards & human factors	— (12 experts)
P11	“I inherently just trust that it works”: Mental Models of DP Libraries Song, Sarathy, Shoemate & Vadhan	2024	CSCW	Standards & human factors	— (5+17 people)



A map of the field

Each bubble is a paper, placed by **year** (left→right) and **research angle** (colour & row); bubble size grows with the paper's length. The field moves from attacks & mechanisms (2022) toward standards & human factors (2026).



What they did ✓ · the gap ✗ · what we fill ●

For every paper: its contribution, the gap it leaves, and how SynthProof closes it.

P1 Stadler et al. 2022

- ✓ **What they did**
First quantitative proof that synthetic data leaks membership for outlier records; open-source attack framework.
- ✗ **The gap**
No signed or checkable release artifact; the DP fix needs a public dataset most curators lack.
- **SynthProof fills it**
A data-blind linter + signed sheet that flags the very leak (domain/seed) they found.

P2 Annamalai et al. 2024

- ✓ **What they did**
Tight empirical auditing of 6 DP synthesizers; black-box MIAs read $\epsilon \approx 0$ even at true $\epsilon = 4$.
- ✗ **The gap**
Audits the mechanism — needs code, model internals & worst-case data; not the shipped document.
- **SynthProof fills it**
Grounds our audit ceiling; explains why our audited $\epsilon = 0.000$ is loose, not 'safe'.

P3 Cebere et al. 2026

- ✓ **What they did**
'Re:cord-play' grey-box audit: 13 privacy violations across 12 DP libraries.
- ✗ **The gap**
Audits library code, not the released artifact/document.
- **SynthProof fills it**
Same 'neighbours + identical randomness' trick as our seed-replay; motivates our 2nd accountant.

P4 Ganev et al. 2025

- ✓ **What they did**
Reading the data domain from the input breaks end-to-end DP → MIA AUC ≈ 1.0 .
- ✗ **The gap**
No artifact field records the domain's provenance so a validator could check it.
- **SynthProof fills it**
External proof of our D1 / domain-source rule; boundary-audit flags data-derived domains.

P5 Ganev et al. 2025

- ✓ **What they did**
Discretization strongly affects utility (+9–44%) and DP; DP domain drops MIA 100% → 50%.
- ✗ **The gap**
Audits discretization; ships no checkable release.
- **SynthProof fills it**
With P4/P6 maps the preprocessing subfield — missing-data handling is the slot we fill.

P6 Ganev et al. 2026

- ✓ **What they did**
SMOTE is inherently non-private; ReconSMOTE reconstructs minority records (precision 1.0).
- ✗ **The gap**
Naive metrics (DCR) miss it; no audit of missing-data handling; no artifact check.
- **SynthProof fills it**
Validates our sanity-gate design; our kNN imputation arm is the SMOTE-like risk we test.

What they did ✓ · the gap ✗ · what we fill ●

For every paper: its contribution, the gap it leaves, and how SynthProof closes it.

P7 Mohapatra et al. 2024

- ✓ **What they did**
First to study DP synthesis with missing data; imputation is ne-expensive; missingness amplifies privacy.
- ✗ **The gap**
Never audits whether the missing-data handling leaks membership.
- **SynthProof fills it**
THE base for our imputation audit — we run the membership check they leave open.

P8 McKenna et al. 2022

- ✓ **What they did**
State-of-the-art marginal synthesizer (up to 118× lower workload error than MST).
- ✗ **The gap**
Ships a model, not a signed/boundary-checked release; reads the domain from data.
- **SynthProof fills it**
The generator we USE; our D1 fix corrects its uncharged use of the private row count.

P9 Abowd et al. 2022

- ✓ **What they did**
Nationwide DP deployment (zCDP + discrete Gaussian); ships 'invariants' outside ϵ .
- ✗ **The gap**
Invariants documented only in prose/policy — not a machine-checkable field.
- **SynthProof fills it**
Our real-world case study; we make 'which fields are outside ϵ ' signed & checkable.

P10 Dibia et al. 2026

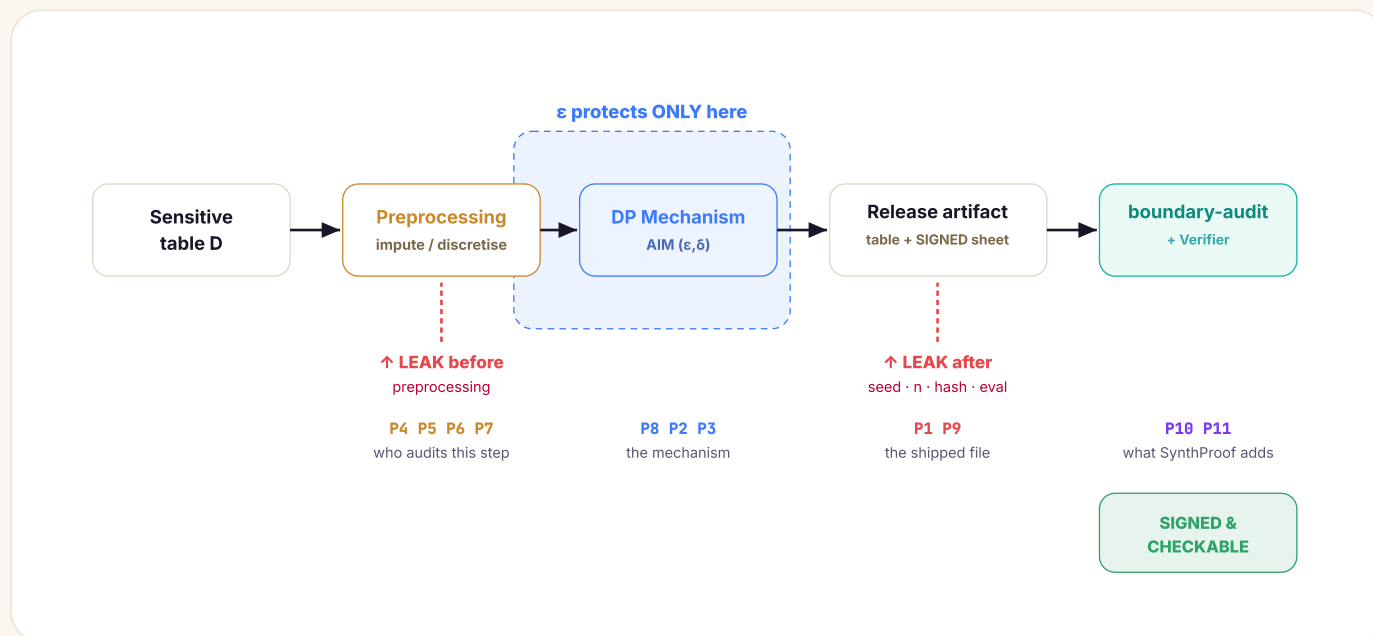
- ✓ **What they did**
Expert-elicited 9-category DP privacy label ($\approx$ our Privacy Data Sheet, field for field).
- ✗ **The gap**
No signing, and no way to report a metric's limits — an expert called it 'privacy theater'.
- **SynthProof fills it**
★ LEAD: we sign the label, make it machine-checkable, and report the audit's operating range.

P11 Song et al. 2024

- ✓ **What they did**
Data analysts trust DP libraries implicitly and do not verify guarantees.
- ✗ **The gap**
Diagnoses blind trust but builds no tool.
- **SynthProof fills it**
★ THE PREMISE: why an automatic, signed, checkable release is needed.

The release boundary — and who guards what

Differential privacy protects only the machine in the middle. A real deployment ships a **file**, and it can leak before the machine (preprocessing) and after it (the label). Each paper studies one stretch of this pipe; SynthProof adds the signed, checkable layer at the end.



Read it like this: P4/P5/P6/P7 audit the preprocessing step · P8/P2/P3 work on the mechanism · P1 & P9 expose what leaks in the shipped artifact · P10 & P11 show the field agreed *what* to disclose but never made it checkable. That last box is SynthProof.



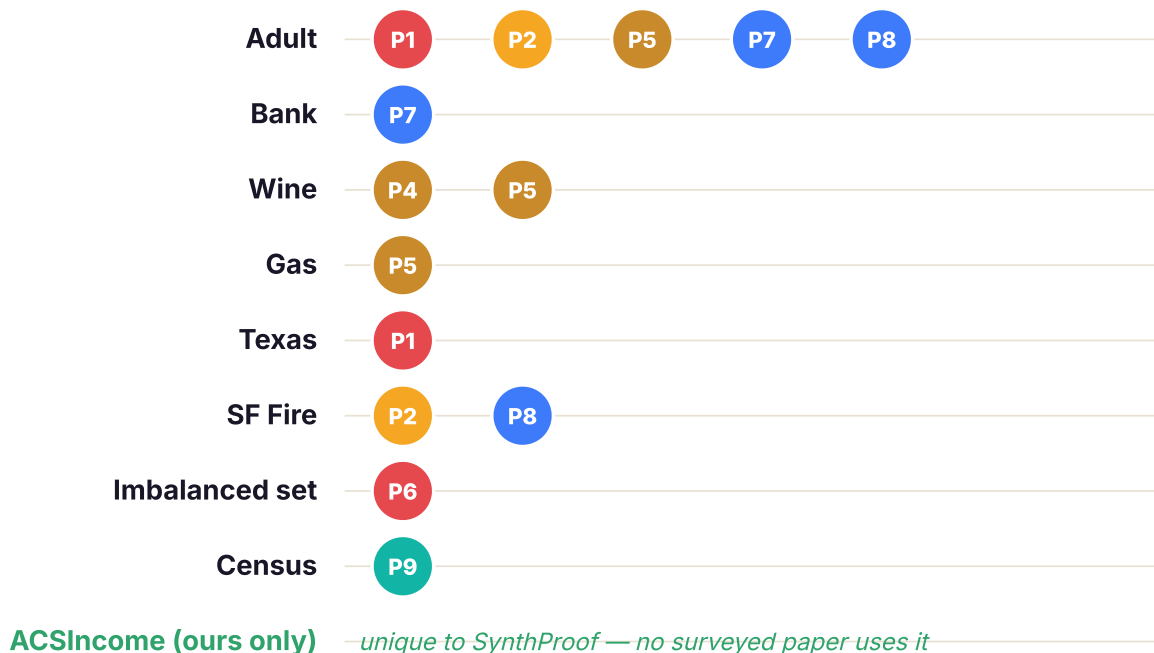
Cross-cutting themes

A filled cell means the paper touches that theme. Four threads run across the corpus — and the boldest one, “reading structure from private data breaks DP,” is the exact bug our D1 fix and domain-source rule address.

	P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11
Reads structure from private data → breaks DP	✓	✓		✓	✓						
Preprocessing is a leak surface				✓	✓	✓	✓				
Naive / black-box metrics miss leaks		✓				✓					
Membership / reconstruction attack	✓	✓		✓		✓					
Not signed / not machine-checkable (the gap)									✓	✓	✓
Real deployment / proposed standard									✓	✓	

Which paper used which dataset

UCI **Adult** is the shared benchmark (5 papers) — where our same-dataset comparison lives. **P7** also uses **Bank**. **ACSIIncome** is ours alone.



The one-line verdict. The field converged on *what* a DP release should disclose (P10) but never on making it *checkable* — no signing (P10), no limit-reporting (“privacy theater”), invariants in prose only (P9), and analysts who never verify (P11). That gap — **signed + machine-checkable + limit-reporting** — is SynthProof.